A FUNDAMENTAL CLASS MAP FROM THE TOP EXTERIOR POWER OF THE DIFFERENTIALS TO THE SHRIEK MODULE, AN ISOMORPHISM ON THE SMOOTH LOCUS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EA0 (sources/stacks-duality/pinned-original-source.tex line 7845).

Allowed standard prerequisites (editorial).

- The fundamental class on the local complete intersection locus. Same chapter, Lemma labelled lemma-fundamental-class-lci, cited at proof lines 7894-7900. Provides the map on U.
- The shriek module is (S_2) over a Cohen-Macaulay base. Same chapter, Lemma labelled lemma-shriek-over-CM, cited at proof lines 7906-7908. Gives the depth condition needed for the Hartog-type extension.
- Extending a section across a closed subset of codimension at least two. Divisors, Lemma labelled divisors-lemma-depth-2-hartog, cited at proof lines 7908-7910. Extends the map from U to X.

Source reference. Original definition 02IO: sources/stacks-chow/tags.txt, source lines 3307–3307. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 069F: sources/stacks-chow/tags.txt, source lines 8039–8039. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E9Z: sources/stacks-duality/pinned-original-source.tex, source lines 7774–7774. Same-chapter pointer into the author environment lemma at source line 7774; invoked at proof line 7871 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E9V: sources/stacks-duality/pinned-original-source.tex, source lines 5935–5935. Same-chapter pointer into the author environment lemma at source line 5935; invoked at proof line 7880 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E9I: sources/stacks-chow/tags.txt, source lines 17853–17853. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7884 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $r \geq 0$. Assume*

1. *Y is Cohen-Macaulay (Properties, Definition 02IO),*
2. *f factors as $X \to P \to Y$ where the first morphism is an immersion and the second is smooth and proper,*
3. *if $x \in X$ and $\dim(\mathcal{O}_{X,x}) \leq 1$, then f is Koszul at x (More on Morphisms, Definition 069F), and*
4. *if ξ is a generic point of an irreducible component of X, then we have $trdeg_{\kappa(f(\xi))}\kappa(\xi) = r$.*

Then with $\omega_{X/Y} = H^{-r}(f^!\mathcal{O}_Y)$ there is a map

$$\wedge^r \Omega_{X/Y} \longrightarrow \omega_{X/Y}$$

which is an isomorphism on the locus where f is smooth.

Proof. Let $U \subset X$ be the open subscheme over which f is a local complete intersection morphism. Since f has relative dimension r at all generic points by assumption (4) we see that the locally constant function of Lemma 0E9Z is constant with value r and we obtain a map

$$\wedge^r \Omega_{X/Y}|_U = \wedge^r \Omega_{U/Y} \longrightarrow \omega_{U/Y} = \omega_{X/Y}|_U$$

which is an isomorphism in the smooth points of f (this locus is contained in U because a smooth morphism is a local complete intersection morphism). By Lemma 0E9V and the assumption that Y is Cohen-Macaulay the module $\omega_{X/Y}$ is (S_2). Since U contains all the points of codimension 1 by condition (3) and using Divisors, Lemma 0E9I we see that $j_*\omega_{U/Y} = \omega_{X/Y}$. Hence the map over U extends to X and the proof is complete. □